

Python Full Stack Developer Virtual Internship

RAJYAVARDHAN | Roll No: 2410030715

B.Tech Computer Science & Engineering (AI/ML) | IILM University, Greater Noida

Program AICTE – EduSkills Virtual Internship	Domain Python Full Stack Developer
Duration 8 Weeks June – August 2026	Status Completed & Certified

Candidate & Program Overview

Student Profile

Name	RAJYAVARDHAN
Roll No.	2410030715
Degree	B.Tech CSE (AI/ML)
Batch	2026–2027
University	IILM University, Greater Noida

Internship Details

Program	AICTE – EduSkills Virtual Internship
Specialization	Python Full Stack Developer
Duration	8 Weeks (June – August 2026)
Program Type	Industry-Focused Virtual Internship
Offer Letter Ref.	C17Y2026M081452593
Certificate ID	44478c1afe30f54e3097
Status	Completed & Certified

Organization Profile: EduSkills Academy & Problem Context

About EduSkills Academy

Mission

Nation Building Through Skills —
bridging the gap between academia and
industry-ready talent.

Accreditation

ISO 9001:2015 & ISO 20000-1:2018
Certified | AICTE-approved training
partner.

Domains Covered

Full Stack Development, Cloud
Computing, AI/ML, Data Science —
curriculum designed by industry experts.

Methodology

Hands-on, project-based learning with
real-world applications and placement
assistance.

Industry Problem Statement

Engineering students frequently master theoretical programming concepts yet lack practical exposure to building complete, industry-ready web applications. Modern organizations require professionals who can own the **entire development lifecycle** — from responsive UI to scalable backend to database deployment.

- ☐ This internship directly addresses the full-stack skills gap by delivering a structured, project-driven curriculum across all layers of a production application.

Solution: Complete Full-Stack Lifecycle Coverage

- Frontend development with modern web technologies
- Backend logic via Python and Flask
- Database design, integration, and management
- REST API development, testing, and deployment concepts
- Version control and collaborative development workflows



Core Project Objectives

Ten structured learning objectives formed the foundation of the internship, mapping directly to industry-required competencies for a Python Full Stack Developer role.



Technologies & Development Toolchain

Frontend Stack

- **HTML5** – Semantic markup & structure
- **CSS3** – Styling & responsive design
- **JavaScript** – Client-side interactivity & DOM manipulation
- **Bootstrap** – Responsive UI component framework

Backend Stack

- **Python** – Core programming language
- **Flask** – Lightweight, modular web framework
- **Jinja2** – Server-side template rendering
- **REST APIs** – Endpoint design & integration

Database

- **MySQL** – Relational database design
- **CRUD Operations** – Create, Read, Update, Delete
- **Python-MySQL Connector** – Database connectivity

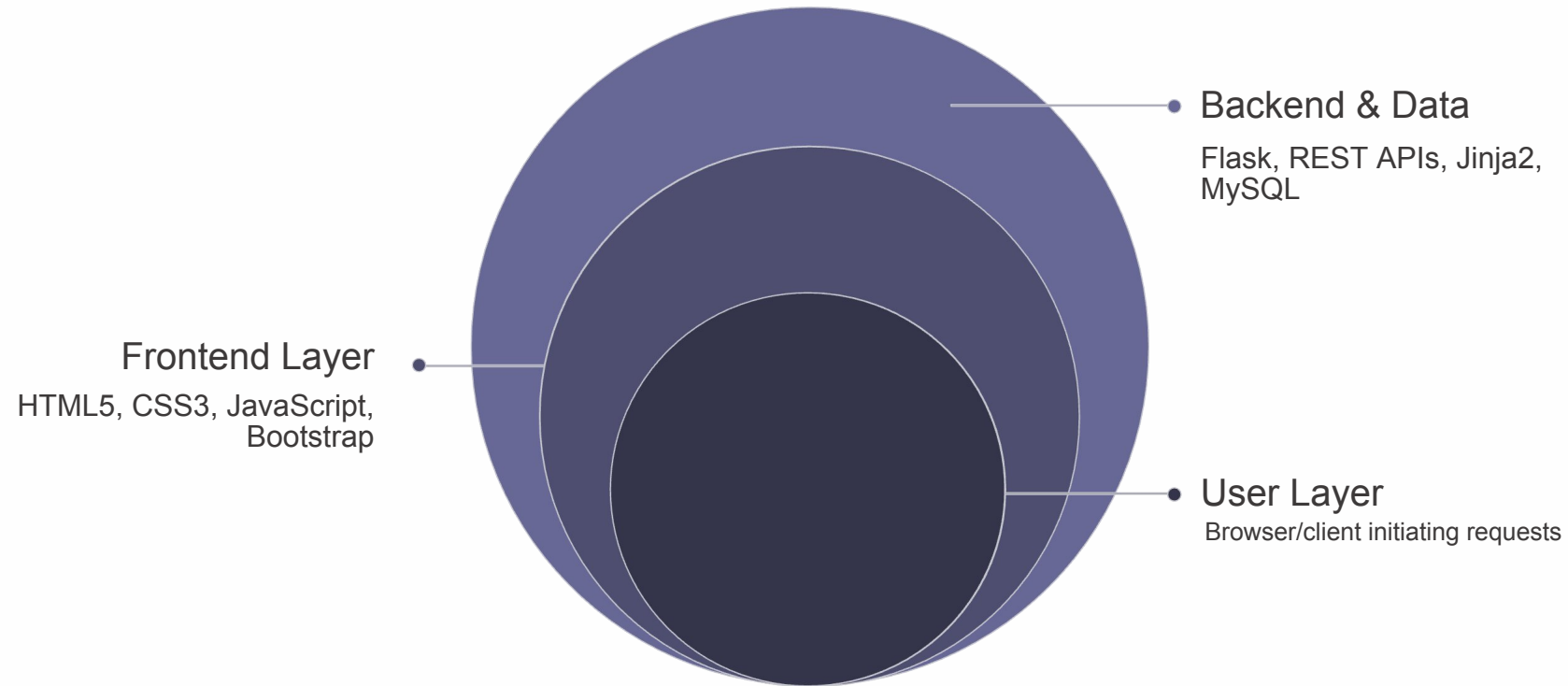
Dev & Deployment Tools

- **VS Code** – Integrated development environment
- **Git & GitHub** – Version control & collaboration
- **Browser DevTools** – Frontend debugging
- **Local Deployment** – Testing & validation



Full Stack Application Architecture

The internship project follows a classic multi-tier architecture, with clear separation of concerns across five distinct layers — from browser-facing UI to persistent data storage.



Data Flow — Request Cycle

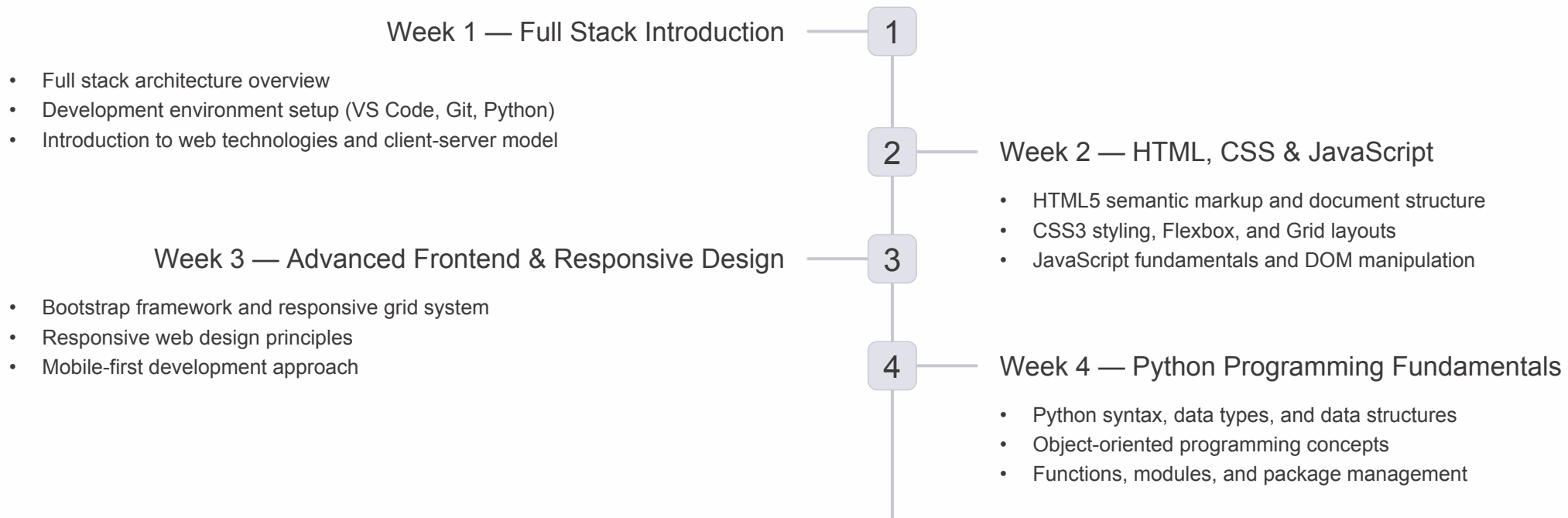
1. User interacts with the **Frontend** (HTML/CSS/JS/Bootstrap)
2. HTTP request routed to the **Flask Backend**
3. **REST API endpoints** process business logic
4. Flask queries the **MySQL Database** via Python connector
5. Response rendered via **Jinja2 templates** back to browser

Design Principles Applied

- **Separation of Concerns** — distinct frontend/backend/data layers
- **RESTful Design** — stateless API communication
- **Responsive UI** — mobile-first Bootstrap grid
- **Secure Auth** — session-based user authentication
- **Modular Code** — Flask Blueprints pattern

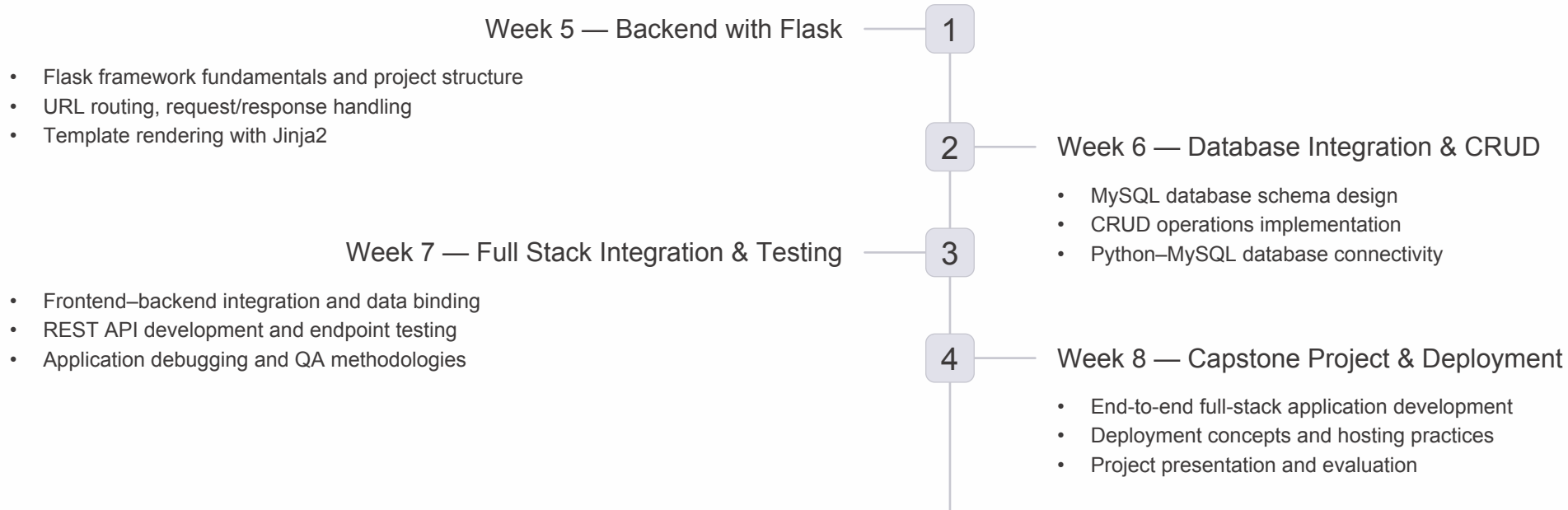
Curriculum Timeline: Weeks 1 – 4

The first four weeks established foundational competencies across the full frontend stack and introduced Python as the backend language.



Curriculum Timeline: Weeks 5 – 8

The second phase advanced into backend engineering, database management, full-stack integration, and culminated in the capstone project submission.



📄 All 8 weeks completed successfully. Capstone project evaluated and certification awarded upon completion of Week 8 deliverables.

Capstone Project: Python Full Stack Web Application

Project Overview

Developed a dynamic, production-grade full-stack web application demonstrating complete end-to-end integration of frontend, backend, and database layers — the culminating deliverable of the 8-week internship program.

Key Features Implemented

- 1

Responsive UI

Modern, mobile-friendly interface built with Bootstrap and custom CSS3
- 2

Authentication System

Secure user login, registration, and session management
- 3

Full CRUD Functionality

Create, Read, Update, Delete operations with MySQL integration
- 4

REST API Endpoints

Structured API layer with proper request handling and validation
- 5

Error Handling & Testing

Systematic form validation, error feedback, and QA debugging

Technology Stack Summary

Layer	Technology
Frontend	HTML5, CSS3, JavaScript
UI Framework	Bootstrap
Backend	Python, Flask
Templating	Jinja2
Database	MySQL
Version Control	Git & GitHub
IDE	VS Code
Deployment	Local / Hosted Server

Key Skills & Demonstrated Learning Outcomes

Eight core competencies were demonstrated and assessed across the internship, each mapped to industry-standard full-stack developer requirements.



Responsive UI Development

Created modern, mobile-friendly user interfaces using HTML5, CSS3, JavaScript, and Bootstrap responsive grid.



Python Backend Development

Built scalable, modular backend logic using Flask framework with Jinja2 templating and route management.



Database Management

Designed and managed MySQL relational databases; implemented full CRUD operations with Python connectivity.



REST API Integration

Developed and integrated RESTful API endpoints; applied stateless request-response communication patterns.



Authentication & Validation

Implemented secure user authentication, session handling, form validation, and input sanitization.



Git & GitHub Workflow

Mastered version control, branching strategies, commit discipline, and collaborative repository management.



Debugging & Testing

Applied systematic debugging using browser DevTools and backend logging; implemented application-level QA.



Full Stack Project Delivery

Completed end-to-end capstone application demonstrating integrated frontend, backend, and database deployment.

Certification & Verification Details

Official Credential Record

Student Name	RAJYAVARDHAN
Roll Number	2410030715
Program	Python Full Stack Developer
Issuing Body	AICTE – EduSkills Academy
Certificate ID	44478c1afe30f54e3097
Offer Letter Ref.	C17Y2026M081452593
Duration	8 Weeks June – August 2026
Accreditation	ISO 9001:2015 & ISO 20000-1:2018
Status	COMPLETED & CERTIFIED

Program Summary & Institutional Endorsement

This internship was conducted under the AICTE–EduSkills virtual internship framework, recognized by the All India Council for Technical Education as a credit-eligible, industry-aligned training program for engineering students.

Key Takeaways for Evaluation

Industry Readiness

Hands-on exposure to production-grade full-stack development workflows and toolchains.

Capstone Delivered

End-to-end web application built and evaluated — demonstrating applied competency across all 10 objectives.

Verified Credentials

Certificate ID and Offer Letter reference available for institutional verification via EduSkills portal.

Certificate & Offer Letter



PRESENTED TO

Rajyavardhan .*has successfully achieved the Internship Credential by completing a 8-week program in the domain of***Python Full Stack***During the internship, the learner demonstrated proficiency in:*

- HTML: Foundations of Web Development
- CSS: Styling and Designing Web Pages & Bootstrap: Responsive Web Design Framework
- JavaScript Fundamentals and DOM Manipulation
- jQuery: Simplifying JavaScript Development
- Python: Programming Fundamentals and Applications
- Building Web Applications Using Django
- SQL: Database Querying and Management
- Git & Version Control: Managing Code Effectively

This certification recognizes the successful completion of the internship requirements.



Issued on: Jul 29, 2026
Certificate ID: 2026-7B1659F7D5


Mito Swain
General Manager
(Learning & Development)
EduSkills

Ref No: C17Y2026M081452593

Internship Offer Letter

Student Name : Rajyavardhan .
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Python Fullstack Developer
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6989ebf52c9a81770646517

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Thank You!



Rajyavardhan

Roll No: 2410030715



**School of Computer Science &
Engineering**

IILM University, Greater Noida, U.P.